Ani Abrahamyan

(463) 210-4664 | aniabrahamyan800@gmail.com | Noblesville, IN | LinkedIn | Portfolio

Education

Purdue University Aug 2025 – Expected May 2029

B.S. Computer Engineering, Minors in Math & Philosophy | GPA: 3.54/4.0 *West Lafayette, IN*

- **Relevant Coursework:** ECE 264 (Advanced C Programming), VIP 27920 (STARS Chip Design Skills)
- **Awards and Honors:** Dean's List, Gilbert J. Easley Scholarship in ECE, InnovateHer Finance Track 1st Place

Professional Experience

STARS Chip Design Team Member Jun 2026 – Aug 2026

Summer Training, Awareness, and Readiness for Semiconductors, Purdue SoCET *Remote*

- Completed the chip design track of Purdue's summer semiconductor program for first- and second-year students
- **Digital design:** Built a memory copier in SystemVerilog from a 4-state Moore FSM controller, data register, and flex counter, verifying transfers with self-checking Verilator testbenches and GTKWave waveforms
- **Synthesis:** Synthesized RTL with Yosys using the FreePDK45 cell library, generating a netlist and finding that flip-flops used 55% of an FSM's area
- **HW/SW integration:** Designed a timing-safe popcount accelerator that counts all 32 bits in parallel, offloading a RISC-V core over AHB with a C driver and achieving a 4.9x speedup over a software implementation

Solution Engineer Co-Op Jun 2026 – Present

Cummins Inc. *Columbus, IN*

- Engineering an AI agent pipeline in Python on Databricks that uses Anthropic's Claude (Sonnet 5) to harmonize inconsistent Cummins master data, starting with engine families, to replace manual data steward review
- Implemented a feedback loop that logs each manual correction to a table and feeds it back into the agent's prompt to reduce repeated errors on ambiguous records
- Built a test framework scoring agent output against ground-truth values, achieving 90% accuracy across 1,800+ records of engine family data ahead of scaling to new data domains

Project Experience

Single-Cycle RISC-V CPU | *Verilog/SystemVerilog, Verilator, RISC-V* GitHub

- Building a single-cycle processor in SystemVerilog for a subset of the 32-bit RISC-V ISA, with a datapath, control unit, register file, ALU, and branch logic, as the base for a 5-stage pipeline extension
- Verifying with self-checking Verilator testbenches and SystemVerilog assertions, targeting edge cases such as signed comparison at the overflow boundary

Systolic Array for 2D Matrix Multiplication | *Verilog/SystemVerilog, Verilator* GitHub

- Designing an output-stationary systolic array for N×N matrix multiplication, the core AI/ML operation; implemented a parameterized multiply-accumulate processing element and now building its Verilator testbench

Extracurricular Activities

Social Media & Outreach Chair May 2026 – Present

Purdue Armenian Student Association *West Lafayette, IN*

- Found the Purdue Armenian Student Association and established it as a constituent chapter of All-ASA, the national network of Armenian Student Associations
- Leading social media and campus outreach; representing Purdue in All-ASA senate meetings and coordinating events with East Coast & Midwest chapters

Sponsorship Organizer Dec 2025 – May 2026

StarkHacks *West Lafayette, IN*

- Conducted outreach and coordinated sponsorships (AMD, Ford) for a student-run hardware hackathon at Purdue, bringing 750+ students from 80 universities and 10 countries to build robotics and AI hardware projects

Technical Skills

Languages: Verilog/SystemVerilog, C, Python, MATLAB
Hardware Design: RTL Design, Computer Architecture, Computer Arithmetic, Digital Logic, RISC-V, FSM Design
Verification: Self-Checking Testbenches, SystemVerilog Assertions, Verilator, GTKWave, Waveform Debugging
Tools: Git, Linux (CLI), Yosys, LTSpice (circuit simulation), Databricks, Power BI